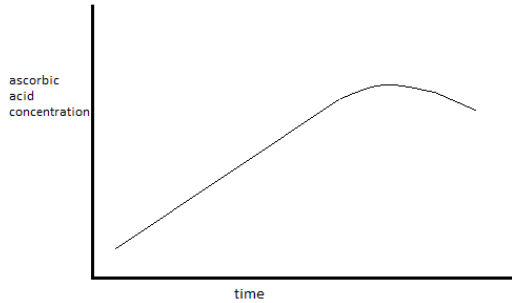


|          |                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(a)(i)  | <p>any <b>one</b> from:</p> <ol style="list-style-type: none"> <li>1 <i>idea that</i> size (of apple) may not be, related / proportional, to ripeness ;</li> <li>2 <i>idea that</i> (individual) apples may, ripen at different rates / AW ;</li> <li>3 <i>idea that</i> other factors may affect, ripeness of apples / diameter of apples ;</li> </ol>                                                          | <b>1</b> |
| 1(a)(ii) | <p>any <b>one</b> from:</p> <ol style="list-style-type: none"> <li>1 (pick apples with the) same / similar / AW, sugar content / colour / softness / age ;</li> <li>2 test / measure / AW, sugar content / colour / softness / age (of apples) ;</li> </ol>                                                                                                                                                      | <b>1</b> |
| 1(b)     | <ol style="list-style-type: none"> <li>1 correct axis labels ;</li> <li>2 correct shaped line ;</li> </ol>  <p>The graph shows a line that starts at a low point on the y-axis, rises linearly to a peak, and then slightly declines. The y-axis is labeled 'ascorbic acid concentration' and the x-axis is labeled 'time'.</p> | <b>2</b> |
| 1(c)(i)  | <ol style="list-style-type: none"> <li>1 90 ;</li> <li>2 50 ;</li> </ol>                                                                                                                                                                                                                                                                                                                                         | <b>2</b> |

| Question        | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks                                                   |      |            |       |                            |                               |        |         |                                                         |                 |        |                    |   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------|------------|-------|----------------------------|-------------------------------|--------|---------|---------------------------------------------------------|-----------------|--------|--------------------|---|
| 1(c)(ii)        | <p>any <b>nine</b> from:</p> <ol style="list-style-type: none"> <li>1 idea of test on at least 5 days ;</li> <li>2 use apples from same, tree / variety / type ;</li> <li>3 use, same / stated, storage conditions (of apples) ;</li> </ol> <p>prepare apple extract solutions:</p> <ol style="list-style-type: none"> <li>4 remove the, skin / seeds, of the apple ;</li> <li>5 use, set / stated, mass of apple ;</li> <li>6 method to, grind / AW, apple ;</li> <li>7 centrifuge / filter ;</li> </ol> <p>measure ascorbic acid concentration using DCPIP:</p> <ol style="list-style-type: none"> <li>8 place, set / stated, volume of (apple extract) solution in, conical flask / AW ;</li> <li>9 idea of adding DCPIP dropwise to (apple extract) solution until blue colour remains ;</li> <li>10 shake / mix / AW (after each drop of DCPIP) ;</li> <li>11 measure / record, volume of DCPIP used (to reach end point) ;</li> <li>12 read volume from bottom of meniscus (in burette) ;</li> <li>13 use at least three replicates for each day <u>and</u> calculate a mean (DCPIP volume) ;</li> <li>14 safety comment with hazard <u>and</u> risk <u>and</u> precaution ;</li> </ol> <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>DCPIP</td><td>irritant / toxin / allergy</td><td>gloves / goggles / mask / PPE</td></tr> <tr> <td>apples</td><td>allergy</td><td>gloves / goggles / mask / PPE / medication / do not eat</td></tr> <tr> <td>knife / scalpel</td><td>injury</td><td>cut away from hand</td></tr> </tbody> </table> | hazard                                                  | risk | precaution | DCPIP | irritant / toxin / allergy | gloves / goggles / mask / PPE | apples | allergy | gloves / goggles / mask / PPE / medication / do not eat | knife / scalpel | injury | cut away from hand | 9 |
| hazard          | risk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | precaution                                              |      |            |       |                            |                               |        |         |                                                         |                 |        |                    |   |
| DCPIP           | irritant / toxin / allergy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | gloves / goggles / mask / PPE                           |      |            |       |                            |                               |        |         |                                                         |                 |        |                    |   |
| apples          | allergy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | gloves / goggles / mask / PPE / medication / do not eat |      |            |       |                            |                               |        |         |                                                         |                 |        |                    |   |
| knife / scalpel | injury                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | cut away from hand                                      |      |            |       |                            |                               |        |         |                                                         |                 |        |                    |   |
| 1(d)            | <p>any <b>two</b> from:</p> <ol style="list-style-type: none"> <li>1 data is continuous ;</li> <li>2 scatter diagram, suggests a linear, relationship ;</li> <li>3 data is normally distributed ;</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                                                       |      |            |       |                            |                               |        |         |                                                         |                 |        |                    |   |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(e)     | <p>any <b>three</b> from:</p> <p><i>for the conclusion:</i></p> <ol style="list-style-type: none"> <li>1 apples have the second lowest <math>\beta</math>-carotene concentration (of the fruits shown) ;</li> <li>2 quoted data from Fig. 1.3 to support ;</li> <li>3 <i>idea that</i> spread of data for apple is lower (than most fruits that were tested, as SD is smaller)</li> </ol> <p><i>against the conclusion:</i></p> <ol style="list-style-type: none"> <li>4 <i>idea that</i> need to calculate, SE / 95% CI (to make a conclusion) ;</li> <li>5 no statistical test carried out ;</li> <li>6 <i>idea that</i> only 5 fruits were tested ;</li> </ol> | 3     |

| Question | Answer | Marks |
|----------|--------|-------|
|----------|--------|-------|